BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 008 · Character sets: ASCII & Unicode

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which data type should store a telephone number beginning with 0? Explain briefly.

Q2. Convert 41_{16} to denary and eight-bit binary.

Q3. Apply AND 00001111 to 10100101. Explain which bits the mask preserves.

Q4. In 6+4 two’s-complement floating-point format, subtract 010000 0010 from 011000 0010. Give a normalised result.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — Character set

Recall the term — Code point

Recall the term — Encoding

Recall the term — ASCII

Recall the term — Unicode

Recall the term — UTF-8

Recall the term — Control character

New learning

Characters need agreed codes — A character set defines characters and their assigned numeric codes. An encoding specifies how those codes are represented as bytes.

Standard ASCII is limited — Seven bits provide $2^7 = 128$ codes for letters, digits, punctuation and control characters. — Eight-bit extensions can provide 256 values, but different extensions may map the extra codes differently.

Unicode supports a much larger repertoire — Unicode assigns code points to characters from many writing systems and symbols. Its first 128 code points match ASCII.

Use the encoding stated in storage questions — Storage depends on the number of code points and the number of bytes used for each one. Do not assume every Unicode character uses the same width. — CODE:
'0' in ASCII = 48 = 00110000_2
integer 0 = 0

Worked example

Encode the text “AB”

1. Start with the character codes — CODE:
A = 65
B = 66 — ASCII assigns a numeric code to each character.

2. Write the 7-bit ASCII values — CODE:
A = 1000001
B = 1000010

3. Add a leading zero — CODE:
A → 01000001
B → 01000010

AB → 01000001 01000010 — The leading zero lets each 7-bit ASCII code sit inside an 8-bit byte; it does not expand the standard ASCII repertoire.

We Do · Practise with guidance

2. Character versus quantity

Use the I Do method and build the answer before checking.

Build the answer

3. Use a code range

For lowercase c : 97 + 2 = 99; uppercase C = 99 − 32 = 67.

4. Choose an encoding

Both ends must interpret the bytes using the same encoding.

5. Calculate storage from stated widths

OCR JUNE 2022 · Q5(e)(ii–iii) · [3]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Question 1 · ADAPTED OCR PROGRAMMING TASK · [3]
Complete the three gaps to count uppercase ASCII letters. asc returns a character's code. A = 65 and Z = 90. The loop examines each character once.
CODE:

```
function countCapitals(text)
count = 0
for each character in text
code = asc(character)
if code >= 65 _____ code <= 90 then
count = _____
endif
next character
return _____
endfunction
```

Adapted from OCR June 2022 Q5(e)(i): simplified loop and changed gaps; classroom allocation of one mark per gap. OCR June 2022, Q5(e), page 24

Question 2

Question 2 · ADAPTED EXAM COMPARISON · [4]
A school app must store English and Chinese messages. Explain why standard ASCII is unsuitable [1], why Unicode is suitable [1], give one similarity between them [1], and explain why the receiver must know the encoding [1].
Contextual adaptation of OCR June 2022 Q5(e)(ii–iii); classroom marks.

Question 3

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Question 3 · ORIGINAL EXAM-STYLE CALCULATIONS · [6]
Use A = 65, a = 97 and 0 = 48. Show a code calculation and an eight-bit byte for each: (a) C, (b) c, (c) the digit character 5. [2 each]

Question 4

Question 4 · ORIGINAL CAPACITY QUESTION · [3]
State how many codes standard seven-bit ASCII provides [1]. Calculate the minimum number of bits needed for a fixed-width character set containing 300 distinct codes, showing why the next smaller width is insufficient. [2]

Question 5

Question 5 · ORIGINAL ENCODING APPLICATION · [4]
A UTF-8 string contains 20 ASCII code points, three euro-sign code points and two emoji code points. The byte widths are 1, 3 and 4 respectively. Ignore metadata. Calculate bytes [2] and bits [1]. Explain why multiplying the number of code points by 2 bytes is not valid here. [1]

Exit Ticket · Answer before checking

Standard ASCII provides 128 character codes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. String: preserve the leading zero; it is an identifier, not a quantity for arithmetic.

Q2. 65; 01000001.

Q3. 00000101. The low four bits are preserved; the high four are cleared.

Q4. 3 − 2 = 1 → 010000 0001.

You Do pages · Independent questions

Q1. CODE:
Gap 1: AND
Gap 2: count + 1
Gap 3: count
One mark each. Trace check: AaZ! has two uppercase letters. OR would incorrectly accept codes outside the uppercase range.

Q2. ASCII lacks Chinese characters (1). Unicode includes the required writing systems (1). Both assign numeric codes to characters (1). The receiver needs the encoding to interpret the byte sequences as the intended characters (1).

Q3. CODE:
C: 65 + 2 = 67 → 01000011
c: 97 + 2 = 99 → 01100011
'5': 48 + 5 = 53 → 00110101
For each: code calculation (1), correct byte (1).

Q4. 128 (1). Eight bits provide 256 codes, which is insufficient (1); nine bits provide 512, so the minimum is 9 (1).

Q5. CODE:
(20 × 1) + (3 × 3) + (2 × 4) = 37 bytes
37 × 8 = 296 bits
Correct weighted calculation (1), 37 bytes (1), 296 bits (1). UTF-8 has variable byte widths; the supplied groups do not all use two bytes (1).

Exit Ticket · Model answer

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

True. Standard ASCII has 7 bits and therefore 128 possible codes.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: